AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM358
Machine Learning & Artificial intelligence

Midterm Examination (MTA)
Spring Semester 2025/2026

Date:15/ April /2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 120 minutes

Instructions:

- Total Marks: **60**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part -I: Question 1: Choose the best correct answer of all the following: (15 Marks)

1 ..... knowledge deals with factoid questions

A Instructional
B Declaritive
C Procedural
D All of the above

2 ..... is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators

A A machine
B A computer
C An Agent
D None of the above

3 ..... sensors measure range based on physical contact, and can be deployed only for sensing objects very close to the robot.

A Range finders
B All of the above
C Tactile sensors
D Sonar sensors

4 ..... involves a second ground receiver with known location, providing millimeter accuracy under ideal conditions.

A Differential GPS
B Range finders
C GPS
D Sonar sensors

5 ..... are physically anchored to their workplace, for example in a factory assembly line or on the International Space Station.

A Manipulators
B mobile manipulator
C Mobile robot
D None of the above

6 .....means "Does the strategy find the optimal solution?"

A Completeness
B Optimality
C Time complexity
D Space complexity

7 ..... search from both the start and the goal, in parallel

A backward search
B Bidirectional search
C forward search
D None of the above

8 ..... means ": How much memory is needed to perform the search?"

A Time complexity
B Optimality
C Space complexity
D Completeness

9 ..... is more concerned with obtaining the correct response to an input stimulus without 'looking inside the box' to see if parts of the mechanism can be associated with discrete real world objects.

A Subsymbolic
B Algebric
C Symbolic
D None of the above

10 ..... finds shallowest solution because always works on shallowest nodes first

A A*
B Best first
C BFS
D DFS

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Part-II: Answer all of the following questions (45 Marks).

Question 2: (15 marks)

A. Discuss the limitations of uninformed search methods in terms of completeness, optimality, and space requirements. Explain how these limitations impact the effectiveness of these methods. [5 marks]

B. A tree can be defined by two vectors. The first vector is the parents vector -s. The second vector is the children vector -t. Each entry in t matrix is a child to the corresponding parent in the same index in the s matrix. Assuming s, t in order for a given tree as:

s	A	A	A	B	C	D	D	D	E	F	F	G	G	H	H	I	I
t	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R

Given the tree, apply BFS to traverse the tree showing all agenda content and traced paths until reaching the goal state if exist assuming that the goal state is N

Question 3: (15 marks)

A. How does pruning affect the number of nodes that need to be examined in a game tree search? [5 marks]

B. Having the following XO sequence of states a long with their values

O X X - O - X O -	0.500
O X X - O - X O X	0.512
O X X - O O X O X	0.561
O X X X O O X O X	0.500

Assume a learning rate of 0.99 what will be updated values adopting gradient-based state value update with each move.

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Question 4: (15 marks)

A. Having the following routing table at node 5

	N34	N61	N2	N9	N54
N92	0.35	0.13	0.24	0.21	0.31
N72	0.09	0.30	0.26	0.01	0.21
N33	0.32	0.40	0.25	0.09	0.12
N85	0.24	0.17	0.25	0.69	0.36

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Having started at node 54 and reaching node 5 via node 72.Assuming it has been in the network for 4.486911 time units, what effect will it have on the routing table on node 5 assume a and b equals 0.070000 and 0.010000 in order for ABC model

B. Given the following search tree for *the stone game(4,2)*, use expecti-minimax algorithm to calculate the value of each node hence decide the winner for that game.



Assume the following switching probabilities for min player:

Action	p
F[1]	0.61
G[0]	0.39
D[2]	0.13
E[1]	0.87

End of Questions

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